Original Research

Telemonitoring of Type 2 Diabetes Mellitus in Italy

Claudio Dario, MD,[1] Roberto Toffanin, MD,[2] Francesco Calcaterra, MD,[2] Claudio Saccavini, PhD,[1] Panagiotis Stafylas, MD, MSc, PhD,[3] Silvia Mancin, MEng,[1] and Elena Vio, MEng[1]

[1]Arsenàl.IT, Veneto's Research Centre for eHealth Innovation, Treviso, Italy.
[2]Local Health Authority of Alto Vicentino, Thiene-Schio, Italy.
[3]HIM SA, Brussels, Belgium.

Abstract

Background: *Type 2 diabetes mellitus (DM) affects 382 million people worldwide.* ***Introduction:*** *This study aimed at assessing whether telemonitoring (TM) of DM patients improves health-related quality of life (HRQoL).* ***Materials and Methods:*** *As part of the RENEWING HEALTH project, 299 DM patients with HbA1c >7.0% were enrolled in a randomized controlled trial, with 208 patients in the TM group and 91 patients in the usual-care group. TM electronically transmitted glucose measurements to physicians during a 12-month follow-up. The SF-36v2 questionnaire was used to assess HRQoL.* ***Results:*** *In a total of 243 patients analyzed, the study did not identify any clinically important improvement in HRQoL, our primary endpoint. There was no statistically significant difference in HbA1c between the two groups; however, outpatient visits and planned hospitalizations were significantly reduced in the TM group ($p < 0.0001$ and $p = 0.02$).* ***Discussion:*** *The results regarding HRQoL might be, at least in part, an artifact stemming from the criteria used to select patients. TM reduced ambulatory visits and planned hospital admissions, an important result that plausibly reflects the fact that clinicians can strictly monitor their patients' health status without face-to-face contacts.* ***Conclusions:*** *Enhancement of HRQoL should represent the most critical goal of DM healthcare delivery. Effects of TM on HRQoL of diabetic patients should be studied further.*

Key words: *blood glucose monitoring, telemonitoring, telemedicine, chronic disease management, diabetes mellitus, health technology assessment*

Introduction

Type 2 diabetes mellitus (DM) affects 382 million people worldwide.[1] DM incidence and prevalence are increasing at an alarming rate. DM takes its heavy toll in both reducing duration of life and incapacitating patients. According to the World Health Organization,[2] from 2004 to 2030, DM will shift from the 19th to the 10th cause of lost Disability-Adjusted Life Years (DALYs). In Italy, more than 3 million people (5.5% of the population) suffer from diabetes and its prevalence augments with age reaching 20.3% among people 75 years or older.[3] If the growth rate continues at current levels, no health system will be able to guarantee equitable access to appropriate healthcare services to chronic patients. As a result, there is a need for innovative strategies, leading to an efficient and socially acceptable approach to healthcare services delivery. Lifestyle changes and/or glucose-lowering medications can keep glucose and HbA1C levels in the normal range and, therefore, favorably change the natural history of DM.[4] Poor adherence to therapy in DM patients represents a key obstacle toward reducing vascular complications; a systematic review of adherence to oral hypoglycemic agents showed that retrospective adherence ranged from 36% to 93% in patients remaining on treatment for 6–24 months.[5] In addition, more people with disabilities live longer, because mortality rates decrease much faster than rates of years lived with disability.[6] As a result, the nonfatal dimension of diseases represents a fundamental perspective for societies, health policy-makers, clinicians, and social scientists.

The rationale behind telemonitoring (TM) adoption comprises its potential role in improving adherence to treatment. Moreover, as observed by Noel[7] and Darkins,[8] telemedicine can have a positive impact on chronic patients by improving quality of services and allocation and use of healthcare resources. TM can contribute to the delivery of more timely and appropriate DM care regardless of patients' location. Thanks to remote monitoring of critical parameters, therapy and its effects can be checked continuously and, when appropriate, adjusted, possibly slowing down DM progression and the insurgence of complications. The European Commission has recently recognized the benefits of TM in chronic disease management, and

particularly in DM, in terms of improvement of health-related quality of life (HRQoL), healthcare quality, and resources use, and it has endorsed its adoption among member states.[9]

HRQL refers to an individual's satisfaction with areas of life that concern or are affected by health, including factors under the purview of healthcare providers and healthcare systems.[10] HRQoL refers to patients' perception of their health status and well-being in terms of physical, psychological, and social functioning. HRQoL reflects the growing emphasis on patients' perspective, beyond disease and illness, toward the positive aspects of health, such as adaptive functioning, autonomy, and personal growth.[11] HRQoL reflects the prominence of patients' perspective and embraces a notion of health status that is broader than that traditionally adopted by a paternalistic view of medicine. From an ethical and existential point of view, patients' perception has irreversibly gained equal legitimacy to that of physicians.[12] HRQoL is now the "gold standard" in the evaluation of healthcare services and outcome assessment.[13] As a result, the central goal of chronic care is not only to slow down progression of pathological conditions and prevent acute worsening but also to preserve and, possibly, improve HRQoL.

The aim of the RENEWING HEALTH initiative, a European project focused on the evaluation of telemedicine services for patients with chronic diseases, was to determine the effects of TM on HRQoL across chronic patients affected by different conditions, specifically DM, CHF, and COPD. A comparison of TM effects across different diseases implied the adoption of a general instrument to measure HRQoL.

In our conceptual framework, TM, our exposure, influences HRQoL, our primary outcome, through a set of intermediate variables. TM allows to closely follow up patients, monitoring their clinical parameters even daily, and to promptly inform them when measurements appear outside an acceptable range. TM also offers both patients and clinicians the opportunity to adopt appropriate decisions early on, before the situation worsens. TM consents to better anticipate and respond earlier to exacerbations, and to learn about factors influencing the condition's progression. TM also offers social support through phone calls to patients who are frequently isolated, encouraging them and plausibly making them feel less lonely and better assisted. TM might improve management of chronic conditions and, as a result, HRQoL, also by decreasing severity of symptoms, anxiety and depression, and improving physical and social functioning and perception of one's own health. Better care might also reduce patients' use of healthcare services, including emergency and hospital care. In essence, the rationale of our study is that TM allows the patient–clinician dyad to become more resilient in dealing with chronic conditions, translating into a better HRQoL.

Methods

This RCT was conceived as part of the REgioNs of Europe WorkINg toGether for HEALTH (RENEWING HEALTH) project, an initiative co-founded by the European Commission and 21 partners from nine European regions, where TM was already standard practice for chronic patients' care.[14] RENEWING HEALTH aimed at evaluating TM services for chronic patients using a rigorous assessment approach, called Model for Assessment of Telemedicine (MAST), enrolling about 7,000 patients.[15] This RCT was carried out within the Local Health Authority (LHA) of Alto Vicentino, in the Veneto Region, northern Italy, without major changes to the existing organization.

The purpose of this pragmatic, parallel, two-arm, open (unblinded) RCT was to assess whether a TM system for DM patients can improve HRQoL.[16] The null hypothesis was that there was no difference between the use or no use of a TM system on DM patients' HRQoL, as measured by the questionnaire SF 36v2. The follow-up period was 12 months. The trial was registered at ClinicalTrials.gov with the identifier NCT01569893. The ethics committee of Vicenza LHA approved the study.

PATIENTS' SELECTION AND RANDOMIZATION

Patients with a diagnosis of DM type 2 already confirmed and HbA1c > 53 mmol/mol (7.0% according to NGSP) were enrolled in the study. Exclusion criteria included: severe comorbidities with life expectancy <12 months, pregnancy, being cognitively unable to participate, and impossibility or inability to use the devices provided or to complete questionnaires in the native language. Participants were enrolled during planned specialist visits; they were informed of the nature and goals of the study. Patients accepting to be involved in the study were assigned a unique ID. Randomization was carried out through a centralized online system that associated the assigned ID to either the intervention or the control group, following the randomization table generated by PASS 2008[17–19] in a proportion of 2:1. The number of patients enrolled in the intervention group was the double than that in the control group, because we wanted to study TM also from an organizational viewpoint. The assumption was a moderate effect size of 0.5 on the primary outcome and a dropout rate of 20%, with power >0.95.

INTERVENTION AND USUAL CARE

The patients randomized to the TM group received telehealth service. Patients were equipped at home with a glucometer and a gateway for data transmission to a Regional eHealth Center (ReHC). According to personalized TM plans devised by a clinician at enrolment, patients measured their blood glucose level. The ReHC processed clinical data automatically to identify possible alarm values, based on thresholds established by specialists

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for each patient. ReHC's operators detected anomalies and informed clinicians when alarms were classified as true. Following the notification by the ReHC's operator, clinicians checked patients' data through a Home Care portal and took appropriate action. Irrespective of the alarm, clinicians could access the Home Care portal any time. In case of an emergency, triggered by a critical glucose concentration, the operator, after confirming the alarm, contacted the patient's relatives and/or the Emergency Department (ED). Participants in the UC group had access to usual care and shared the paper log books of glucometer tests with their general practitioner or specialist during planned visits.

In addition, both the target and the control group received a telecare system for detection of emergency situations at patients' homes, such as falls. The telecare device allowed patients to contact the ReHC 24/7 through an emergency button. Moreover, operators periodically checked patients' general status by phone, regardless of the diagnosis and the severity of the condition.

OUTCOMES

The primary outcome was HRQoL, as measured by the SF-36v2 questionnaire[20] at the beginning and at the end of the follow-up period (12 months). The secondary outcomes, measured at the start and at the conclusion of the follow-up period, included HbA1c concentration, outpatient and ER visits or procedures with a diabetologist, hospitalization rates (overall and due to DM), bed days of hospital care (overall and due to DM), and mental status measured through the Hospital Anxiety and Depression Scale (HADS).[21]

DATA COLLECTION, EVALUATION, AND STATISTICAL ANALYSIS

Normality of data was tested using the Shapiro–Wilk test. Healthcare resources data were over-dispersed and not normally distributed, so groups were compared using the Mann–Whitney test. Differences between groups are shown as differences of medians and 95% confidence intervals. For rates, differences between groups were characterized by risk ratios with a 95% confidence interval and tested by the Exact Poisson test.

For SF36 and HADS results, a *t*-test or a Wilcoxon Mann–Whitney test was used to compare groups, depending on whether data were or not normally distributed. Data are displayed by means and standard deviations; however, for not normally distributed variables, data are also expressed in terms of medians and interquartile ranges. Differences between groups and a 95% confidence interval are referred to as means or medians, depending on the test used.

We analyzed data on an intention-to-treat basis. Therefore, if data were available, we included in the analysis both compliant and noncompliant patients. However, we made no attempt to impute "missing data" from those participants who were lost to follow-up.[22] For patients who died, data were available, but obviously the SF 36 questionnaire at the end of the follow-up could not be administered. For patients lost to follow-up, for reasons other than death, clinical data and the SF-36 questionnaire at the end of the follow-up were not available, because they had withdrawn their informed consent to participate in the study. For the intervention group, we excluded from the analysis those patients who had withdrawn before the data collection started. The analysis was carried out using software R, version 3.0.1.

PATIENTS' FLOW

In this study, 299 patients were enrolled (208 in the TM group and 91 in the UC group) from March to November 2012. *Figure 1* describes the reasons that some patients were not enrolled and others dropped out. Two hundred forty three patients were analyzed for the primary outcome (166 patients in the TM group and 77 in the UC group).

Results

Mean age was about 73 years, and mean HbA1c was 7.94%. There were no significant differences between the two groups in any of the baseline clinical and demographic characteristics (*Tables 1* and *2*). About half of the patients in both groups were treated with insulin, with a mean daily dose of 35.76 versus 38.46 IU ($p > 0.05$) for the TM and the UC groups, respectively. More than half of the patients suffered from heart conditions.

The results of the study are presented in *Table 3*. A nonsignificant improvement of the Physical Component Summary (PCS) score was observed in the TM group (0.09; −0.11 to 2.61; $p = 0.07$). Contrary to expectations, the Mental Component Summary (MCS) score improved in the UC group (−4.49; −5.37 to −1.55, $p < 0.001$). Statistical significance was detected regarding the following subscales: bodily pain, general health, vitality, social functioning, and mental health. However, this statistical difference, together with other variations within scales and subscales, is not clinically significant.[23] In addition, correlation analysis between HbA1c and both PCS and MCS, and between HbA1c and both anxiety and depression were carried out. R^2 was 0.01 for the correlation between HbA1c–MCS, HbA1c–anxiety, and HbA1c–depression; R^2 was 0.04 for the correlation between HbA1c and PCS.

Regarding the consumption of healthcare resources, a significant reduction in the number of outpatient visits to diabetologists (total and with procedures) ($p < 0.001$) was observed in the TM group. Planned hospitalization rates were also statistically lower in the intervention group ($p = 0.02$). No other statistically significant differences in the use of healthcare

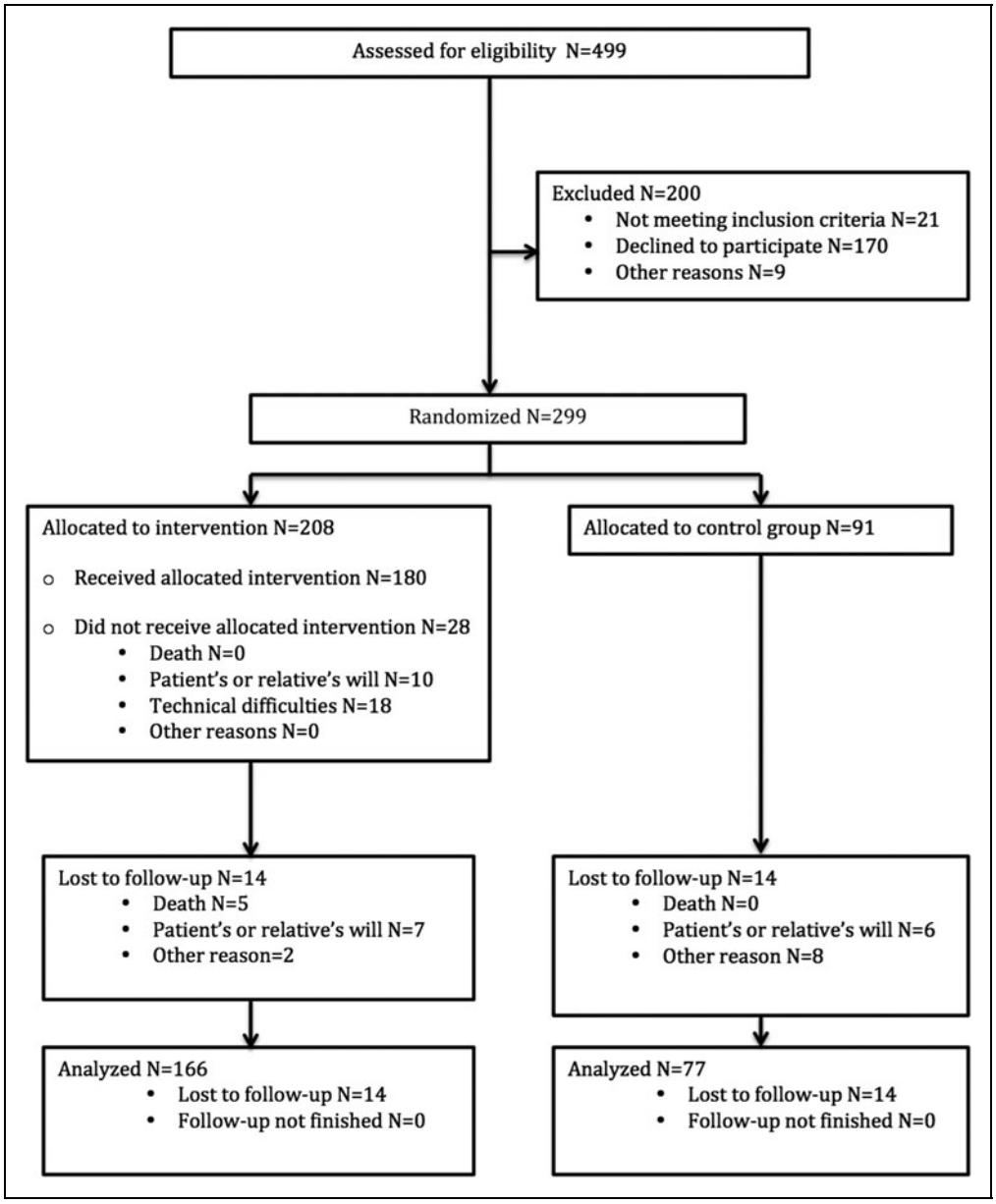

Fig. 1. Participants' flow.

resources in the two groups were observed ($p > 0.05$). The risk of death was higher in patients of the TM group than in those of the UC group, but it did not reach statistical significance ($p = 0.33$).

Discussion

Several studies show that the adoption of TM services improves control of glucose metabolism.[24] TM can also reduce hospitalizations and emergency departments' and specialty-care visits, contributing to significant cost reduction.[25,26] Moreover, TM can improve adherence to therapy, which represents one of the main problems faced by chronic patients and by those affected by DM in particular. It is still unclear whether TM can improve HRQoL.[27] The main aim of our RCT was to determine whether the adoption of TM for DM patients improved their quality of life, and reduced anxiety, depression, HbA1c, outpatient and ED visits, and hospitalizations.

This study showed a significant reduction of outpatient visits and planned hospitalizations; however, such reduction did not translate into better HRQoL, our primary endpoint. Nonetheless, the discouraging findings might be, at least in part, due to:

1. The selection of patients, by and large, with a satisfactory glucose control (>7%);
2. The lack of correlation between the biochemical parameter (HbA1c) used to select patients and the primary endpoint (HRQoL) and a secondary endpoint (HADS); and
3. The use of a generic tool (SF-36v2) to measure HRQoL.

The diagnosis of type 2 DM is based on the measurement of biochemical parameters, specifically fasting or random blood glucose, HbA1c, and oral glucose tolerance test.[28] Our patients were selected on the basis of an HbA1c level above 7%, regardless of their HRQoL. Most of them had a reasonably good glycemic control, also in view of their age. In fact, HbA1c distribution was heavily skewed to the left (median = 7.6%, range 7.0–12.0%) and only 15% of patients (36/243) had poor diabetic control, that is, an HbA1c level above 9.0%.[29] The 2015 ADA recommendations suggest a target for HbA1c of <7% in patients with short duration of DM, long life expectancy, and no significant cardiovascular disease. However, in patients with limited life expectancy, advanced micro- or macro-vascular complications, extensive comorbid conditions,

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Table 1. Main Baseline Data

MEASUREMENT	TM GROUP	UC GROUP	*p* VALUE
Sample size (*n*)	208	91	
Age (mean in years)[a] (SD)	73.05 (5.79)	73.04 (5.28)	0.89 ([W])
Men (%)	119 (57%)	49 (53%)	0.57
Height (mean in cm)[a] (SD)	168.23 (8.39)	167.88 (9.20)	0.75[b]
Weight (mean in kg)[a] (SD)	81.78 (15.68)	79.08 (14.08)	0. 27 ([W])
Years from first diagnosis[a]	15.01 (10.24)	16.01 (9.84)	0.29 ([W])
Familiar history with diabetes	128 (61.84%)	58 (63.04%)	0.62
HbAc1 (%) (SD)[a]	7.94 (0.98)	7.93 (1.10)	0.59 ([W])
Dietary therapy	45 (21.26%)	21 (23.91%)	0.76
Use of oral antidiabetic	158 (75.85%)	60 (66.30%)	0.09
Biguanides: metformin	135 (85.35%)	52 (86.89%)	0.94
Glitazone: pioglitazone	13 (8.28%)	4 (6.56%)	0.78
Alpha-glucosidase inhibitors	8 (5.10%)	2 (3.28%)	0.73
Glinidi: Repaglinide	13 (8.28%)	2 (3.28%)	0.24
DPP-4 inhibitors	16 (10.19%)	4 (6.56%)	0.60
Incretin mimetics	3 (1.91%)	0 (0.00%)	0.56
Use of insulin	104 (50.24%)	50 (54.35%)	0.53
Dose of insulin (mean IU)[a] (SD)	35.76 (22.90)	38.46 (23.16)	0.44 ([W])
Main symptom			
None	126 (60.58%)	63 (69.23%)	0.19
Polyuria	43 (20.67%)	16 (17.58%)	0.65
Polydipsia	21 (10.10%)	3 (3.30%)	0.06
Weight loss	38 (18.27%)	17 (18.68%)	0.99
Other	1 (0.48%)	0 (0.00%)	0.99
Comorbidity			
Heart diseases	116 (56.04%)	54 (58.70%)	0.76
Stroke	0 (0.00%)	1 (1.09%)	0.31
Dementia	0 (0.00%)	1 (1.09%)	0.31
COPD	13 (6.28%)	6 (6.52%)	0.99
Connective tissue disease/rheumatic disease	6 (2.90%)	5 (5.43%)	0.32
Ulcer disease	1 (0.48%)	0 (0.00%)	0.99
Liver disease	3 (1.45%)	3 (3.26%)	0.38

Table 1. *continued*

MEASUREMENT	TM GROUP	UC GROUP	*p* VALUE
Hemiplegia	1 (0.48%)	3 (3.26%)	0.09
Renal disease	16 (7.73%)	9 (9.78%)	0.72
Cancer	0 (0.00%)	0 (0.00%)	NA
Other	29 (14.01%)	9 (9.78%)	0.41
Alcohol and tobacco consumption			
Alcohol			0.39
Daily	45 (26.32%)	13 (16.25%)	
5–6 days/week	6 (3.51%)	4 (5.00%)	
3–4 days/week	13 (7.60%)	6 (7.50%)	
1–3 days/month	30 (17.54%)	15 (18.75%)	
Less than once/month	74 (42.69%)	39 (50.00%)	
Smoking			
Ever smoker (%)	50/172 (29.07%)	17/77 (22.08%)	0.11
Smoker	8 (4.68%)	0 (0.00%)	
Never smoker	122 (70.76%)	60 (76.25%)	
Former smoker	42 (24.56%)	17 (21.25%)	
Smoker			
Years of smoking[a]	30.63 (15.50)	0 (0.00)	NA
Number of cigarettes[a] (mean/day)	7.25 (6.40)	0 (0.00)	NA.
Former smoker			
Years of smoking[a]	25.79 (11.69)	23.24 (10.42)	0.59 ([W])
Number of cigarettes[a] (mean/day)	17.19 (12.15)	20.88 (12.97)	0.32 ([W])
Baseline data			
SF 36v2 ([W])			
PCS[a]	43.94 (9.96)	43.76 (11.29)	0.99
MCS[a]	48.35 (9.70)	45.17 (10.88)	0.03
HADS([W])			
Anxiety[a]	4.27 (3.06)	5.13 (4.11)	0.31
Depression[a]	4.03 (3.25)	4.81 (4.18)	0.33

[a]Mean (SD).

[b]*t*-test.

(w) Mann–Whitney test. If not specified, we used chi-squared test or Fisher test (depending on frequencies in the cells).

Table 2. Additional Baseline Data

DEMOGRAPHIC and SOCIAL/ECONOMIC AND OTHER DATA			
MEASUREMENT	TM GROUP N (%)	UC GROUP N (%)	*p* VALUE
Education			0.99
No formal schooling	3 (1.75%)	0 (0.00%)	
Less than primary school	8 (4.68%)	2 (2.50%)	
Primary school	113 (65.50%)	53 (67.50%)	
Secondary school	29 (16.96%)	7 (8.75%)	
High school	16 (9.36%)	15 (18.75%)	
College/University	3 (1.75%)	2 (2.50%)	
Postgraduate degree	0 (0.00%)	0 (0.00%)	
Marital status			0.59
Never married	5 (2.92%)	3 (3.75%)	
Currently married	129 (75.44%)	55 (68.75%)	
Separated	0 (0.00%)	1 (1.25%)	
Divorces	1 (0.58%)	0 (0.00%)	
Widowed	35 (19.88%)	19 (25.00%)	
Cohabitating	2 (1.17%)	1 (1.25%)	
Employment status			0.45
Government employee	4 (2.34%)	2 (2.50%)	
Nongovernment employee	40 (23.39%)	19 (23.75%)	
Self-employed	6 (3.51%)	5 (6.25%)	
Nonpaid	0 (0.00%)	0 (0.00%)	
Student	0 (0.00%)	0 (0.00%)	
Homemaker	6 (3.51%)	5 (6.25%)	
Retired	116 (67.25%)	47 (60.00%)	
Unemployed (able to work)	0 (0.00%)	1 (1.25%)	
Unemployed (unable to work)	0 (0.00%)	0 (0.00%)	
PC use	25 (14.62%)	17 (21.25%)	0.21
Mobile phone use	127 (73.68%)	59 (75.00%)	0.99
Is there at least an adult who lives with you?			0.16
Yes	143 (83.63%)	63 (78.75%)	
Is there anyone who assists you at home?			0.41
No	104 (60.23%)	42 (53.75%)	
Relative	65 (38.01%)	32 (40.00%)	
Caregiver	2 (1.17%)	2 (2.50%)	
Private nurse	0 (0.00%)	0 (0.00%)	

Table 2. *continued*

DEMOGRAPHIC and SOCIAL/ECONOMIC AND OTHER DATA			
MEASUREMENT	TM GROUP N (%)	UC GROUP N (%)	*p* VALUE
Integrated home care	0 (0.00%)	0 (0.00%)	
Other	1 (0.58%)	3 (3.75%)	
What means of transportation do you use to travel to the health facility?			0.23
Walking/cycling	3 (1.17%)	3 (5.00%)	
Public transport	7 (4.09%)	2 (2.50%)	
Private transport	161 (94.15%)	74 (92.50%)	
Ambulance	1 (0.58%)	0 (0.00%)	
Is there anyone who accompanies you to the health facility?			0.32
Yes	101 (59.06%)	53 (66.25%)	

a history of severe hypoglycemia, or long-standing DM, a less stringent HbA1c target, that is, <8%, may well be appropriate.[28] Results from large RCT even show that aggressive reduction of HbA1c management in high-risk patients might increase mortality without diminishing the risk of major cardiovascular events.[30]

The poor results in terms of HRQoL improvement in this RCT might also derive from the weak correlation between HbA1c and HRQoL.[27] Patients with similar glycated hemoglobin levels experience wide variation in functionality and perception of health status.[31] To put it in another way, HbA1c concentration levels above 7.0% do not automatically translate into an impairment of HRQoL.[32] Our sample of DM patients at enrollment displays a lack of correlation between HbA1c and both PCS and MCS (R^2 is 0.04 and 0.01, respectively).

A similar difficulty derives from the good enough affective status of most patients in our sample and the poor correlation between HbA1c and measures of anxiety and depression. The HADS scale categorizes individuals into four groups: normal (0–7 score), mild (8–10), moderate (11–14), and severe (15–21). In our sample, at enrollment, the mean scores for anxiety and depression were 4.54 and 4.28, respectively. Using the cut-offs 8+ and 15+, the percentages of patients with elevated anxiety or depression in the total sample were 19% and 17%, and 2% and 1%, respectively. These scores are similar to those of the general population.[33] The sample of DM patients at enrollment displays a lack of correlation between HbA1c and both

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Table 3. Study Results

OUTCOME		TM GROUP BASELINE	TM GROUP 12 MONTHS	UC GROUP BASELINE	UC GROUP 12 MONTHS	MEAN DIFFERENCES (±SD) / MEDIAN DIFFERENCES (CI)	*p* VALUE
HRQoL with SF36 v.2 (TM = 166, UC = 77)							
PCS	Mean ± SD	44.17 ± 9.91	45 ± 5.41[a]	43.82 ± 11.11	43.44 ± 6.08[a]	1.21	
	Median (IR)	45.07 (37.00–52.99)	45.51 (41.09–49.83)	42.22 (34.56–54.77)	42.57 (38.38–49.43)	0.09 (–0.11 to 2.61)	0.07
MCS	Mean ± SD	48.48 ± 9.58	47.76 ± 3.07[a]	45.11 ± 10.88	47.9 ± 3.49[a]	–3.51	
	Median (IR)	49.64 (42.07–55.89)	48.14 (45.71–50.14)	44.81 (38.81–52.57)	47.8 (45.88–50.29)	–4.49 (–5.37 to 1.55)	<0.001
Physical function (PF)	Mean ± SD	42.61 ± 12.8	42.56 ± 7.55[a]	40.41 ± 14.55	41.28 ± 8.59[a]	–0.92	
	Median (IR)	44.41 (35.99–54.93)	43.63 (38.66–49.84)	38.09 (31.78–57.03)	39.91 (36.19–51.09)	–2.6 (–4.21 to 0.01)	0.03
Role-physical (RP)	Mean ± SD	42.62 ± 10.69	42.31 ± 3.73[a]	39.46 ± 12.27	40.47 ± 4.29[a]	–1.32	
	Median (IR)	42.16 (37.26–51.96)	42.16 (40.44–45.58)	37.26 (32.36–47.06)	39.7 (37.99–43.13)	–2.44 (–4.03 to 0.74)	0.59
Bodily pain (BP)	Mean ± SD	48.91 ± 10.91	49.84 ± 3.82[a]	49.01 ± 11.52	48.34 ± 4.04[a]	1.6	
	Median (IR)	46.9 (41.41–62.12)	49.13 (47.21–46.96)	49.44 (41.41–62.12)	48.49 (45.67–52.93)	3.18 (1.54–4.65)	0.02
General health (GH)	Mean ± SD	42.86 ± 9.04	44.51 ± 2.87[a]	43.7 ± 8.9	43.69 ± 2.83[a]	1.66 (–0.01 to 3.31)	0.05
Vitality (VT)	Mean ± SD	51.64 ± 9.31	50.63 ± 3.38[a]	49.78 ± 10.16	51.31 ± 3.7[a]	–2.54 (–4.25 to 0.82)	0.004
Social functioning (SF)	Mean ± SD	48.83 ± 8.18	49.71 ± 1.79[a]	47.08 ± 10.85	49.48 ± 2.38[a]	–1.52	
	Median (IR)	51.4 (45.94–56.85)	50.28 (49.08–51.48)	51.4 (40.49–56.85)	50.43 (48.03–51.63)	–0.15 (–0.150 to 0.149)	<0.001
Role-emotional (RE)	Mean ± SD	42.2 ± 11.13	42.08 ± 3.82[a]	38.11 ± 13.05	40.48 ± 4.49[a]	–2.49	
	Median (IR)	44.22 (32.56–51.99)	42.77 (38.77–44.44)	40.33 (28.67–48.1)	41.24 (37.23–43.91)	–2.36 (–4.90 to 0.20)	0.46
Mental health (MH)	Mean ± SD	47.87 ± 10.34	46.64 ± 4.02[a]	44.67 ± 10.81	46.75 ± 4.2[a]	–3.31 (–5.08 to 1.53)	<0.001
HADS (TM = 166.UC = 77)							
Anxiety	Mean ± SD	4.22 ± 3.05	4.23 ± 3.01	5.12 ± 4.07	3.95 ± 2.68	1.17	0.021
Depression	Mean ± SD	3.92 ± 3.16	3.63 ± 3.14	4.87 ± 4.2	3.82 ± 4.04	0.76	0.158

		TM GROUP	UC GROUP	MEAN DIFFERENCE (±SD) / MEDIAN DIFFERENCE or Rate Ratio (CI)	DELTA	*p* VALUE
HbA1c (TM = 168, UC = 78)						
Difference HbA1c	Mean ± SD	–0.26 ± 0.92	–0.27 ± 0.99	–0.01	–3.70%	
	Median(IR)	–0.2 (–0.7 to 0.1)	–0.25 (–0.8 to 0.1)	0.05 (–0.2 to 0.2)		0.76
Healthcare resources (TM = 172. UC = 79)						
Hospitalizations for all causes	Rate	0.02	0.04	0.67 (0.43–1.08)	–33%	0.08
Hospitalizations related to diabetes	Rate	0.01	0.01	0.99 (0.43–2.48)	–1%	0.99
Planned hospitalizations for all causes	Rate	0.01	0.02	0.53 (0.29–0.95)	–47%	0.02
Emergency hospitalizations for all causes	Rate	0.01	0.01	1.03 (0.47–2.43)	3%	0.99

continued →

Table 3. Study Results *continued*

		TM GROUP	UC GROUP	Mean DIFFERENCES (± SD) Median difference or Rate Ratio (CI)	DELTA	*p* VALUE
Bed days for hospitalized patients	Mean ± SD	21.13 ± 43.12	21.68 ± 31.73	−0.55	−2.5%	
	Median (IR)	13 (1.5–23)	9 (2–23.25)	4 (−7 to 6)		0.99
Bed days for hospitalized patients related to diabetes	Mean ± SD	20.83 ± 16.51	25.4 ± 37.88	−4.57	−18.0%	
	Median (IR)	14 (9.25–35.75)	9 (4–20)	5 (−46 to 30)		0.53
Emergency department visits	Mean ± SD	0.53 ± 1.06	0.29 ± 0.68	0.24	82.8%	
	Median (IR)	0 (0–1)	0 (0–0)	0(-5.14×10^{-5} to 5.97×10^{-5})		0.07
Outpatient visits with procedures with diabetologist	Mean ± SD	1.8 ± 2.03	3 ± 2.67	−1.2	−40.0%	
	Median (IR)	1 (0–3)	3 (1–4)	−2 (−2 to 8.56×10^{-5})		<0.0001
Outpatient visits with diabetologist	Mean ± SD	0.97 ± 1.01	1.43 ± 1.07	−0.46	−32.2%	
	Median (IR)	1 (0–2)	1 (1–2)	0 (−1 to 7.72×10^{-5})		<0.0001
Primary care visits with procedures	Mean ± SD	30.52 ± 18.31	31.89 ± 19.62	−1.37	−4.3%	
	Median (IR)	28 (18–40)	30 (16.5–39.5)	−2 (−5 to 4)		0.79

[a]Calculated adjusting for enrolment values using ANOVA.

anxiety and depression, where R^2 is 0.01 and 0.01, respectively. The lack of correlation between HbA1c–HRQoL and HbA1c–HADS illustrates how most chronic patients adapt to their condition up to the last phases of the natural course, when health status rapidly and irremediably crumbles. Chronic conditions, such as DM, undermine health status perception and weaken confidence in the future, only after decades during which signs and symptoms emerge slowly after physiopathological changes, specifically loss of β-cell function, due to factors such as elevated glucose and lipid levels, inflammation, and oxidative and endoplasmic reticulum stress.[34] Therefore, the fact that the HRQoL and HADS of sampled patients were satisfactory at baseline may explain why TM did not have any effect on HRQoL and HADS.

The use of a generic tool (SF-36v2), instead of a condition-specific instrument, to measure HRQoL might have contributed to our results. Generic tools to measure HRQoL take into account constructs that are applicable to any healthy or sick individual.[35] As a result, they are useful in comparing the effects of interventions across different health status and conditions, which was one of the central goals of the RENEWING HEALTH project.

In our study, TM reduced outpatient visits and planned hospitalizations, plausibly because a between-visits close monitoring of biochemical parameters, together with periodic phone calls by ReHC's operators, allowed clinicians to gain a satisfactory representation of patients' health status. TM provides almost a motion picture of every patient's clinical condition, instead of isolated snapshots supplied by periodic visits. As a consequence, healthcare professionals might feel more in control and reduce the frequency of face-to-face contacts, including planned hospitalizations, that are frequently motivated by the perceived need to stabilize, under strict supervision in a hospital setting, an unbalanced glucose metabolism. In this way, more healthcare resources could be converted for patients with more severe diabetes. Moreover, the reduction of outpatient visits and planned hospitalizations prevents many travels that may create costs and inconvenience to patients with mobility restrictions.

This study contributes toward building evidence regarding the effectiveness of TM as a tool aiding clinicians to detect impending deteriorations, adjust patients' therapy, or activate immediate professional help. Moreover, this research shows that TM can improve healthcare services' efficiency through better management of chronic diseases, diminishing direct costs and also travel times for both health professionals and patients.

Conclusions

When assessing HRQoL, objective health status represents a marginal issue, whereas patients' personal viewpoints on their own well-being becomes the overriding concern.[36] Sustaining a good HRQoL should represent the most critical goal of DM

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healthcare delivery. The statistically significant differences identified by our study in some items of SF-36v2 are not sufficient to maintain that TM improves HRQoL. However, this finding might, to some extent, be explained by the selection criteria, that is, HbA1c >7%, and the poor correlation between HbA1c and quality of life as measured by a generic tool. To capture the potential effect of TM on quality of life, it would have been more convenient to select patients on the basis of functional status. Still, this study contributes toward building further evidence that TM can reduce outpatient visits and planned hospitalizations. These are important results from both an economic and an organizational standpoint, contributing to cost reduction and allowing clinicians to focus attention on the most complex cases.

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No competing financial interests exist.

REFERENCES

1. Guariguata L, Whiting DR, Hambleton I, Beagley J, Linnenkamp U, Shaw JE. Global estimates of diabetes prevalence for 2013 and projections for 2035. *Diabetes Res Clin Pract* **2004;**103,137–149.
2. WHO (2008). The global burden of disease: 2004 update. Available at www.who.int/healthinfo/global_burden_disease/2004_report_update/en/ (last accessed November 17, 2015).
3. ISTAT, *Annuario Statistico Italiano, 2013* (i.e., Italian Statistical Yearbook), Rome, Italy: Istituto nazionale di statistica, 2013.
4. Phillips LS, Ratner RE, Buse JB, Kahn SE. We can change the natural history of type 2 diabetes. *Diabetes Care* **2014;**37:2668–2676.
5. Cramer JA. A systematic review of adherence with medications for diabetes. *Diabetes Care* **2004;**27:1218–1224.
6. Vos T et al. Global, regional, and national incidence, prevalence, and years lived with disability for 301 acute and chronic diseases and injuries in 188 countries, 1990–2013: A systematic analysis for the Global Burden of Disease Study 2013. *Lancet* **2015;**386:743–800.
7. Noel HC, Vogel DC, Erdos JJ, Cornwall D, Levine F. Home telehealth reduces healthcare costs. *Telemed J E Health* **2004;**10:170–183.
8. Darkins A, Ryan P, Kobb R, Foster L, Edmonson E, Wakiefield B, Lancaster A. Care Coordination/Home Telehealth: The systematic implementation of health informatics, home telehealth, and disease management to support the care of veteran patients with chronic conditions, *Telemed J E Health* **2008;**14:1118–1126.
9. Commission of the European Communities, COM (2008) 689 communication from the commission to the european parliament, the council, the european economic and social committee and the committee of the regions on telemedicine for the benefit of patients, healthcare systems and society, 4 November 2008, Brussels, Belgium.
10. Wilson IB, Cleary PD. Linking clinical variables with health-related quality of life. *JAMA* **1995;**1995:59–65.
11. Singer B, Ryff C. *New horizons in health: An integrative approach*. Washington, D.C.: National Academy Press, **2001**.
12. Asadi-Lari M, Tamburini M, Gray D. Patients' needs, satisfaction, and health related quality of life: Towards a comprehensive model. *Health Quality Life Outcome* **2004;**2:32.
13. Spilker B, Revicki D. Taxonomy of Quality of Life. In: Spilker B, ed. *Quality of life and pharmaco economics in clinical trials*. Baltimore: Lippincott Williams & Wilkins, **1996**.
14. Mancin S, Saccavini C, Dario C, Centis G. Telemedicine for the remote monitoring of patients with chronic disease: European project renewing health. *Int J Cars* **2011;**6:S305–S366.
15. Kidholm K, Ekeland AG, Jensen LK, Rasmussen J, Pedersen CD, Bowes A, Flottorp SA, Bech M. A model for assessment of telemedicine applications: Mast. *Int J Technol Assess Health Care* **2012;**28:44–51.
16. Lesaffre E. Superiority, equivalence, and non-inferiority trials. *Bull NYU Hosp Jt Dis* **2008;**66:150–154.
17. Piantadosi S. *Clinical trials: A methodologic perspective*, 2nd ed. New Jersey: John Wiley & Sons, **2005**.
18. Pocock SJ. *Clinical trials—A practical approach*. New York: John Wiley & Sons, **1983**.
19. Rosenberger WF, Lachin JM. *Randomisation in clinical trials—theory and practice*. New York: John Wiley & Sons, **2002**.
20. Ware, JE Jr. SF-36® Health Survey Update, Available at www.sf-36.org/tools/SF36.shtml#LIT (last accessed February 4, 2015).
21. Zigmond AS, Snaith RP. The hospital anxiety and depression scale. *Acta Psychiatr Scand* **1983;**67:361–370.
22. Hollis S, Campbell F. What is meant by intention to treat analysis? Survey of published randomised controlled trials. *BMJ* **1999;**319:670–674.
23. Maruish ME. (Ed) *User's manual for the SF-36v2 Health Survey,* 3rd ed. Lincoln, RI: QualityMetric Incorporated.
24. Paré G, Moqadem K, Pineau G, St-Hilaire C. Clinical Effects of Home Telemonitoring in the Context of Diabetes, Asthma, Heart Failure and Hypertension: A Systematic Review. Agence d'évaluation des technologies et des modes d'intervention en santé, Quebec Department of Health, Montreal, Canada. *J Med Internet Res* **2010;**12:e21.
25. Paré G, Moqadem K, Pineau G St-Hilaire C. Systematic review of the effects of home telemonitoring in the context of diabetes, pulmonary diseases and cardiovascular diseases. *ETMIS* **2009;**5:1–75.
26. Chase HP1, Pearson JA, Wightman C, Roberts MD, Oderberg AD, Garg SK. Modem transmission of glucose values reduces the costs and need for clinic visits. *Diabetes Care* **2003;**26:1475–1479.
27. U.K. Prospective Diabetes Study Group. Quality of life in type 2 diabetic patients is affected by complications but not by intensive policies to improve blood glucose or blood pressure control (UKPDS 37). U.K. Prospective Diabetes Study Group. *Diabetes Care* **1999;**22:1125–1136.
28. American Diabetes Association. Standards of medical care in diabetes –2015. *Diabetes Care* **2015;**38:S1–S93.
29. U.S. Department of Health and Human Services, Health Resources and Services Administration. DIABETES HBA1C {POOR CONTROL}. Available at www.hrsa.gov/quality/toolbox/508pdfs/diabetesmodule.pdf (last accessed September 2, 2015)
30. ACCORD (The Action to Control Cardiovascular Risk in Diabetes Study Group). Effects of Intensive Glucose Lowering in Type 2 Diabetes. *N Engl J Med* **2008;**358:2545–2559.
31. Nielsen AB, et al. The Relationship between HbA1c Level, Symptoms and Self-Rated Health in Type 2 Diabetic Patients. *Scand J Prim Health Care* **2011;**29:157–164.

32. Weinberger M, Kirkman MS, Samsa GP, Cowper PA, Shortliffe EA, Simel DL, Feussner JR. The relationship between glycemic control and health-related quality of life in patients with non-insulin-dependent diabetes mellitus. *Med Care* **1994;**32:1173–1181.

33. Hinz A, Brähler E. Normative values for the hospital anxiety and depression scale (HADS) in the general German population. *J Psychosom Res* **2011;**71:74–78.

34. Butler AE, Janson J, Bonner-Weir S, Ritzel R, Rizza RA, Butler PC. Beta-cell deficit and increased beta-cell apoptosis in humans with type 2 diabetes. *Diabetes* **2003;**52:102–110.

35. Patrick DL, Deyo RA. Generic and disease-specific measures in assessing health status and quality of life. Med Care **1989;**27(Suppl 3): S217–S232.

36. Polonsky WH. Understanding and assessing diabetes-specific quality of life. *Diabetes Spectrum* **2000;**13:36.

Address correspondence to:
Elena Vio, MEng
Arsenàl.IT
Veneto's Research Centre for eHealth Innovation
Viale Guglielmo Oberdan 5 Treviso
Treviso 31100
Italy

E-mail: evio@consorzioarsenal.it; info@consorzioarsenal.it

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